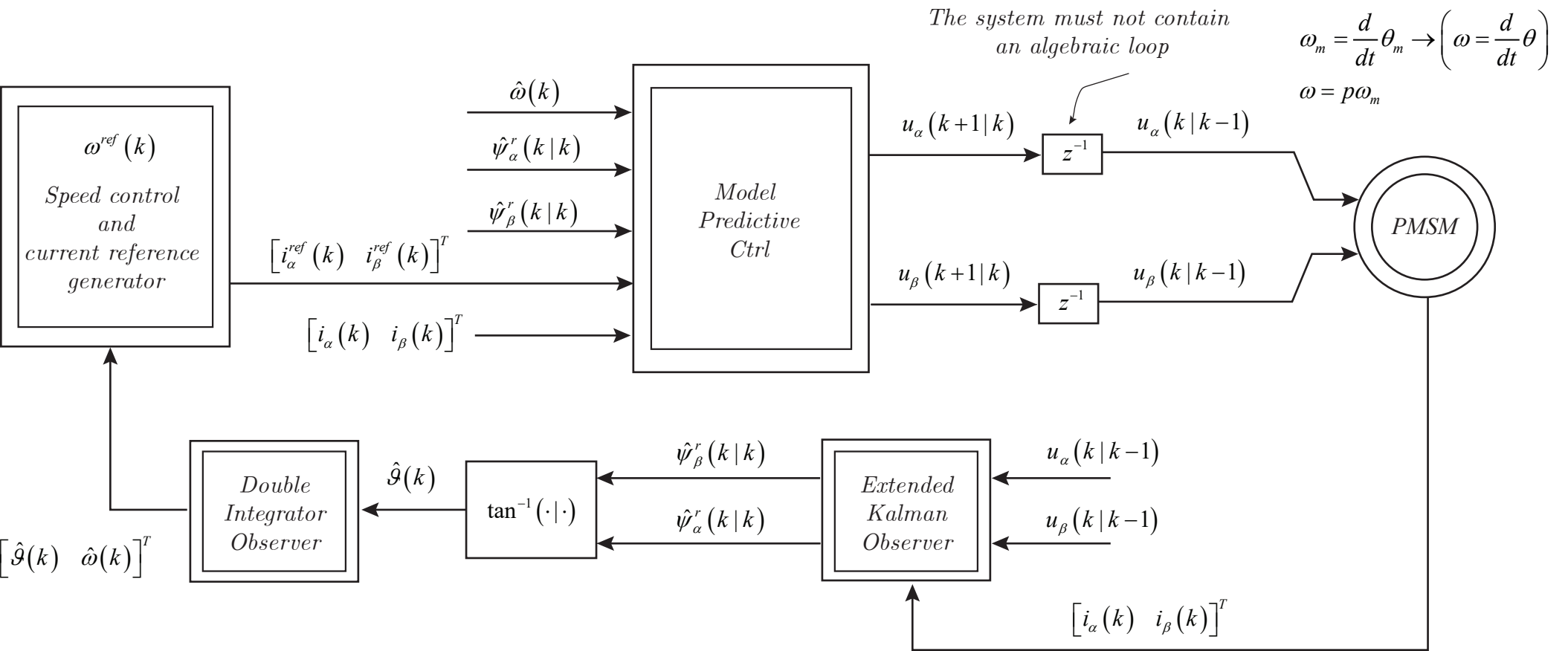

The system must not contain an algebraic loop
$\omega_m = \frac{d}{dt}\theta_m \rightarrow \left(\omega = \frac{d}{dt}\theta\right)$
$\omega = p\omega_m$
$\omega^{ref}(k)$
Speed control and current reference generator
$\hat{\omega}(k)$
$\hat{\psi}_\alpha^r(k|k)$
$\hat{\psi}_\beta^r(k|k)$
$\left[i_\alpha^{ref}(k) \quad i_\beta^{ref}(k)\right]^T$
$\left[i_\alpha(k) \quad i_\beta(k)\right]^T$
Model Predictive Ctrl
$u_\alpha(k+1|k)$
$u_\beta(k+1|k)$
z^{-1}
z^{-1}
$u_\alpha(k|k-1)$
$u_\beta(k|k-1)$
PMSM
$\left[i_\alpha(k) \quad i_\beta(k)\right]^T$
Extended Kalman Observer
$u_\alpha(k|k-1)$
$u_\beta(k|k-1)$
$\hat{\psi}_\beta^r(k|k)$
$\hat{\psi}_\alpha^r(k|k)$
$\tan^{-1}(\cdot|\cdot)$
$\hat{\vartheta}(k)$
Double Integrator Observer
$\left[\hat{\vartheta}(k) \quad \hat{\omega}(k)\right]^T$